

Study of the out-of-equilibrium dynamics of one-dimensional Bose gases

Étude de la dynamique hors équilibre des gaz de Bosons unidimensionnels

04-11-2025

Guillaume THÉMÈZE

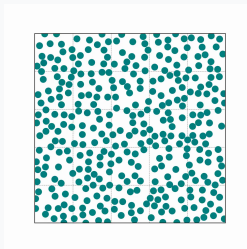
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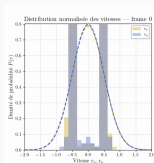
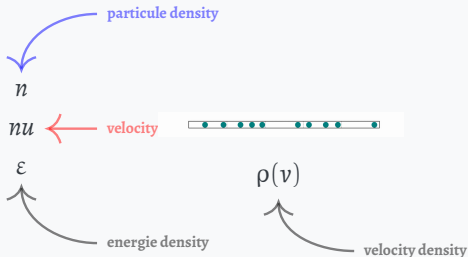
🐙 github.com/THEMEZE

Rapidities and GGE in a classical system

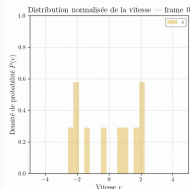
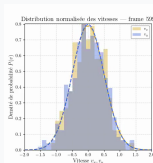
Chaotic/Ergodic



Integrable/non-ergodic



thermalisation
→



Thesis Problematic

Scientific Question

How can we characterize the spatio-temporal evolution of a 1D quantum gas after a perturbation, in terms of the local rapidity distribution and its link to Generalized Gibbs Ensembles (GGE)?



Objectives

- Lieb and Liniger;
- GHD;
- Manip;
- Bipartition;
- Fluctuation

The N-body problem: the scattering phase

Kinetic energy

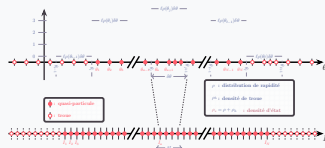
$$\hat{H} = -\frac{\hbar^2}{2m} \sum_{a=1}^N \hat{\partial}_{x_a}^2 + g \sum_{1 \leq a < b \leq N} \delta(\hat{x}_a - \hat{x}_b)$$

1D repulsion strength

Contact potential

$$\psi_{\{\theta_a\}}(x_1 < \dots < x_N) \propto \sum_{\text{perm. } \sigma} e^{i\Phi_\sigma(\{\theta_a\})} e^{i\frac{m}{\hbar} \sum_{b=1}^N \theta_{\sigma(b)} x_b}$$

$$\frac{m}{\hbar} \theta_a L + \sum_{b=1}^N \Phi(\theta_a - \theta_b) = 2\pi I_a \quad a = 1, \dots, N$$



$$\begin{cases} L\rho(\theta)\delta\theta & : \# & \text{rapidities} & \in [\theta, \theta + \delta\theta] \\ L\rho_h(\theta)\delta\theta & : \# & \text{holes} & \in [\theta, \theta + \delta\theta] \\ L\rho_s(\theta)\delta\theta & : \# & \text{states} & \in [\theta, \theta + \delta\theta] \end{cases}$$

The N-body problem: the scattering phase

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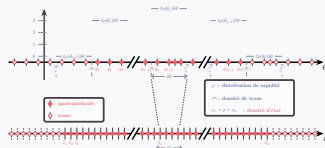
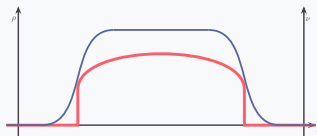
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$\{\theta_1, \dots, \theta_N\}$

$|\{\theta_a\}\rangle$

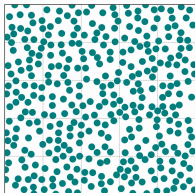
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GHD

Hydrodynamic scale

*Chaotic/Ergodic**Integrable/non-ergodic*

local particle density

$$q_0(x; t) = n(x; t)$$

local velocity mean

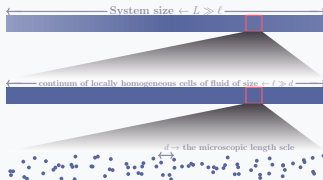
$$q_1(x; t) = nu(x; t)$$
$$q_2(x; t) = \varepsilon(x; t)$$

local energy density



$$\rho(\theta, x; t)$$

local velocity density



Comparaison des commandes

- **Uncover** : espace réservé

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- **Visible** : sans espace réservé

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- **Onslide** : garde l'espace

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- **A** : alterne contenu

A frame with title and subtitle

Subtitle here

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua

Itemized list:

- Lorem ipsum
- Dolor sit amet
 - Consectetur
 - Adipiscing elit
- Sed do eiusmod
 - Tempor incididunt
 - Ut labore et dolore
 - Magna aliqua

A frame with title only

Theorem

$$e^{i\pi} + 1 = 0$$

Démonstration

$$e^{iz} = \cos z + i \sin z$$

therefore

$$\begin{aligned} e^{i\pi} + 1 &= \cos \pi + i \sin \pi + 1 \\ &= -1 + i \times 0 + 1 \\ &= 0 \end{aligned}$$

A frame with background image

You can still add title and subtitle

You can also use a background in the title slide by setting:

```
\frame[plain,bg=demo-background.jpg]{\titlepage}
```

demo-argue1

A plain frame has no headline

<code>\Alegreya</code>	Lorem ipsum dolor sit amet
<code>\AlegreyaMedium</code>	Lorem ipsum dolor sit amet
<code>\AlegreyaExtraBold</code>	Lorem ipsum dolor sit amet
<code>\AlegreyaBlack</code>	Lorem ipsum dolor sit amet
<code>\AlegreyaSansThin</code>	Lorem ipsum dolor sit amet
<code>\AlegreyaSansLight</code>	Lorem ipsum dolor sit amet
<code>\AlegreyaSans</code>	Lorem ipsum dolor sit amet
<code>\AlegreyaSansMedium</code>	Lorem ipsum dolor sit amet
<code>\AlegreyaSansExtraBold</code>	Lorem ipsum dolor sit amet
<code>\AlegreyaSansBlack</code>	Lorem ipsum dolor sit amet

Alert! A *plain* frame does not show the progress bar but still appears in it unless the frame comes after `\End`

A **STANDOUT** frame can be used to focus attention

In combination with *plain*,
it makes a nice thank-you slide!



<https://github.com/piazzai/arguelles>
<https://ctan.org/pkg/beamertheme-arguelles>